Unit 4 – Conditional Probability

Probability, Bayes' Theorem and Probability Distributions

Contents

1. Probability: Foundations

Probability provides a mathematical way to describe uncertainty.

Examples include:

- Will a coin land heads?
- Will a machine fail during the next hour?
- How many defective items will appear in a sample?
- What is the probability that a customer purchases a product?
- What is the probability that a randomly selected student passed an examination?

Before studying conditional probability, Bayes' theorem and probability distributions, we need a precise vocabulary.

1.1 Random Experiment

A **random experiment** is a process whose exact outcome cannot be known beforehand.

Examples:

- tossing a coin,
- rolling a die,
- drawing a card,
- measuring the lifetime of a component,
- counting the number of customers entering a store.

1.2 Sample Space

The **sample space**, usually denoted by S or Ω, is the set of all possible outcomes of a random experiment.

For one coin toss,

$$S = \{H, T\}.$$

For one six-sided die,

$$S = \{1, 2, 3, 4, 5, 6\}.$$

For two coin tosses,

$$S = \{HH, HT, TH, TT\}.$$

1.3 Events

An **event** is a set of outcomes from the sample space.

Suppose

$$S = \{1, 2, 3, 4, 5, 6\}$$

for a die.

Then the event "an even number occurs" is

$$A = \{2, 4, 6\}.$$

The event "the result is greater than 4" is

$$B = \{5, 6\}.$$

1.4 Basic Set Operations in Probability

For events A and B:

$$A \cup B$$

means A **or** B occurs.

$$A \cap B$$

means A **and** B occur.

$$A^c$$

means A does **not** occur.

1.5 Probability Axioms

For every event A,

$$0 \le \mathbb{P}(A) \le 1.$$

The sample space must occur:

$$\mathbb{P}(S) = 1.$$

The impossible event has probability

$$\mathbb{P}(\varnothing) = 0.$$

For mutually exclusive events A and B,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B).$$

In general,

$$\boxed{\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)}$$

because the intersection would otherwise be counted twice.

The complement rule is

$$\boxed{\mathbb{P}(A^c) = 1 - \mathbb{P}(A)}$$

1.6 Illustration

A fair die is rolled once.
Let

$$A = \{\text{even number}\}, \qquad B = \{\text{number greater than } 3\}.$$

Then

$$A = \{2, 4, 6\}$$

and

$$B = \{4, 5, 6\}.$$

Therefore,

$$A \cap B = \{4, 6\}.$$

Hence

$$\mathbb{P}(A) = \frac{3}{6} = \frac{1}{2},$$

$$\mathbb{P}(B) = \frac{3}{6} = \frac{1}{2},$$

and

$$\mathbb{P}(A \cap B) = \frac{2}{6} = \frac{1}{3}.$$

Thus

$$\mathbb{P}(A \cup B) = \frac{1}{2} + \frac{1}{2} - \frac{1}{3} = \boxed{\frac{2}{3}}.$$

2. Classical and Frequentist Probability

There are several ways to interpret probability. Two of the most important are the **classical** and **frequentist** interpretations.

2.1 Classical Probability

Classical probability is appropriate when all possible outcomes are equally likely.

If an experiment has n equally likely outcomes and m outcomes are favourable to event A, then

$$\boxed{\mathbb{P}(A) = \frac{\text{number of favourable outcomes}}{\text{total number of equally likely outcomes}}}$$

or simply

$$\mathbb{P}(A) = \frac{m}{n}.$$

[redacted]

A fair die is rolled. Find the probability of obtaining a number greater than 4.
The sample space is

$$S = \{1, 2, 3, 4, 5, 6\}.$$

The favourable outcomes are

$$A = \{5, 6\}.$$

Therefore,

$$\mathbb{P}(A) = \frac{2}{6} = \boxed{\frac{1}{3}}.$$

2.2 Frequentist Probability

The frequentist interpretation defines probability through repeated experiments.

Suppose an experiment is repeated n times and event A occurs n_A times.

Its relative frequency is

$$\frac{n_A}{n}.$$

As the number of repetitions becomes very large,

$$\boxed{\mathbb{P}(A) \approx \frac{n_A}{n}}$$

and ideally

$$\mathbb{P}(A) = \lim_{n\to\infty} \frac{n_A}{n}.$$

This interpretation is particularly useful when theoretical probabilities are unavailable.

A manufacturing plant observes 5000 components. Of these, 135 are defective.
An empirical estimate of the probability that a randomly selected component is defective is

$$\mathbb{P}(D) \approx \frac{135}{5000} = 0.027.$$

Thus the estimated defect probability is

$$\boxed{0.027}$$

or 2.7%.

2.3 Classical versus Frequentist Interpretation

	Classical	**Frequentist**
Main idea	Count equally likely outcomes	Observe long-run frequency
Typical use	Coins, dice, cards	Experiments and real data
Requires repeated data?	No	Usually yes
Example	Fair die probability 1/6	Observed failure rate

1. Identify the random experiment.
2. Write the sample space if practical.
3. Clearly define the event.
4. Ask whether outcomes are equally likely.
5. If yes, use
$$\mathbb{P}(A) = \frac{\text{favourable outcomes}}{\text{total outcomes}}.$$
6. If probability is estimated from repeated observations, use relative frequency.
7. Simplify and check that the answer lies between 0 and 1.

3. Conditional Probability: Motivation

Ordinary probability describes uncertainty when we know only the original sample space.

Conditional probability asks a different question:

How should a probability change after we learn additional information?

Consider a standard deck of 52 cards.

Let

$$A = \{\text{card is an ace}\}.$$

Before receiving any additional information,

$$\mathbb{P}(A) = \frac{4}{52} = \frac{1}{13}.$$

Now suppose we are told that the card is a spade.

The possible cards are no longer all 52 cards.

The relevant sample space has been reduced to the 13 spades.

Among these 13 cards, only one is an ace.

Therefore,

$$\mathbb{P}(\text{Ace} \mid \text{Spade}) = \frac{1}{13}.$$

Now suppose instead that we are told that the card is a face card or an ace. The information changes the relevant sample space again.

The fundamental idea is:

Conditioning means restricting the sample space to the information we know to be true.

4. Definition of Conditional Probability

For events A and B, provided that

$$\mathbb{P}(B) > 0,$$

the conditional probability of A given B is

$$\boxed{\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}}$$

The notation

$$\mathbb{P}(A \mid B)$$

is read as

"the probability of A, given B."

The event after the vertical bar is the information already known.
Thus:

$$\mathbb{P}(A \mid B)$$

means we know that B occurred and ask how likely A is under this condition.

4.1 Why the Formula Works

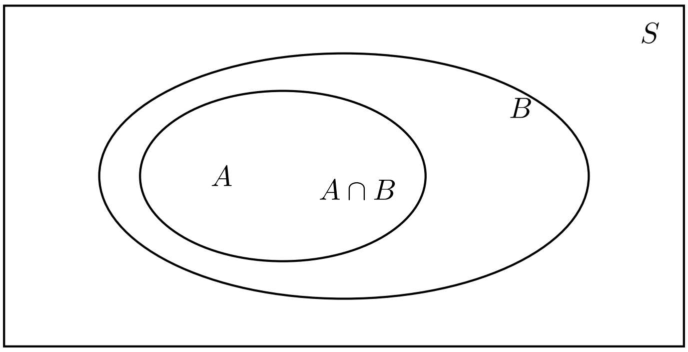
Once B is known to have occurred, only outcomes inside B remain possible.
Among those outcomes, event A occurs precisely in

$$A \cap B.$$

Hence,

$$\mathbb{P}(A \mid B) = \frac{\text{probability belonging to both } A \text{ and } B}{\text{probability belonging to } B}.$$

4.2 Visualisation



When conditioning on B, imagine discarding everything outside B.
Then calculate what proportion of B also lies in A.

4.3 Illustration: Students

In a class of 100 students:

- 60 study mathematics,
- 40 study programming,
- 25 study both.

A student is known to study programming. Find the probability that the student also studies mathematics.
Let

$$M = \{\text{studies mathematics}\}$$

and

$$P = \{\text{studies programming}\}.$$

We require

$$\mathbb{P}(M \mid P).$$

Using the definition,

$$\mathbb{P}(M \mid P) = \frac{\mathbb{P}(M \cap P)}{\mathbb{P}(P)}.$$

From the data,

$$\mathbb{P}(M \cap P) = \frac{25}{100},$$

and

$$\mathbb{P}(P) = \frac{40}{100}.$$

Therefore,

$$\mathbb{P}(M \mid P) = \frac{25/100}{40/100} = \frac{25}{40} = \boxed{\frac{5}{8}}.$$

Notice that the denominator is now 40, not 100, because we already know that the student belongs to the programming group.

4.4 Illustration: Die

A fair die is rolled.
Let

$$A = \{\text{number is even}\}$$

and

$$B = \{\text{number is greater than } 3\}.$$

Then

$$A = \{2, 4, 6\}$$

and

$$B = \{4, 5, 6\}.$$

Their intersection is

$$A \cap B = \{4, 6\}.$$

Hence

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Now,

$$\mathbb{P}(A \cap B) = \frac{2}{6}, \qquad \mathbb{P}(B) = \frac{3}{6}.$$

Therefore,

$$\mathbb{P}(A \mid B) = \frac{2/6}{3/6} = \boxed{\frac{2}{3}}.$$

The same result follows intuitively.
Given that the number exceeds 3, the possible outcomes are only

$$\{4, 5, 6\}.$$

Two of these three outcomes are even.

4.5 Direction Matters

In general,

$$\boxed{\mathbb{P}(A \mid B) \neq \mathbb{P}(B \mid A)}$$

These represent different questions.

For example,

$$\mathbb{P}(\text{student is a programmer} \mid \text{student uses Linux})$$

is not generally the same as

$$\mathbb{P}(\text{student uses Linux} \mid \text{student is a programmer}).$$

4.6 Conditional Probability SOP

When asked for a conditional probability:

1. Translate the wording into
$$\mathbb{P}(A \mid B).$$
2. Identify the event after "given", "among", "knowing that", or "provided that". This becomes B.
3. Write
$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$
4. Find the intersection $A \cap B$.
5. Find the probability of the conditioning event B.
6. Divide.
7. Check that the result is between 0 and 1.
8. Interpret the answer using the reduced sample space.

5. Multiplication Rule

Starting with

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)},$$

multiply both sides by $\mathbb{P}(B)$:

$$\boxed{\mathbb{P}(A \cap B) = \mathbb{P}(B)\mathbb{P}(A \mid B)}$$

Similarly,

$$\boxed{\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A)}$$

Therefore,

$$\boxed{\mathbb{P}(A)\mathbb{P}(B \mid A) = \mathbb{P}(B)\mathbb{P}(A \mid B)}$$

5.1 Sequential Interpretation

The multiplication rule is useful for experiments that happen in stages.

For example:

probability of first event × probability of second event given the first.

5.2 Illustration: Cards without Replacement

Two cards are drawn from a standard deck without replacement.
Find the probability that both cards are aces.
Let

$$A_1 = \{\text{first card is an ace}\}$$

and

$$A_2 = \{\text{second card is an ace}\}.$$

Then

$$\mathbb{P}(A_1) = \frac{4}{52}.$$

If the first card was an ace, only 3 aces remain among 51 cards.
Thus

$$\mathbb{P}(A_2 \mid A_1) = \frac{3}{51}.$$

Therefore,

$$\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(A_1)\mathbb{P}(A_2 \mid A_1)$$

$$= \frac{4}{52} \cdot \frac{3}{51}.$$

Hence

$$\mathbb{P}(A_1 \cap A_2) = \boxed{\frac{1}{221}}.$$

5.3 Chain Rule

For three events,

$$\boxed{\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\mathbb{P}(B \mid A)\mathbb{P}(C \mid A \cap B)}$$

More generally,

$$\mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_n)$$

equals

$$\mathbb{P}(A_1)\mathbb{P}(A_2 \mid A_1)\mathbb{P}(A_3 \mid A_1 \cap A_2) \cdots .$$

6. Independence

Two events are independent if learning that one occurred does not change the probability of the other.

Events A and B are independent if

$$\boxed{\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)}$$

Provided the relevant probabilities are nonzero, this is equivalent to

$$\mathbb{P}(A \mid B) = \mathbb{P}(A)$$

and

$$\mathbb{P}(B \mid A) = \mathbb{P}(B).$$

This gives the intuitive meaning of independence:

knowledge of one event provides no information about the other.

6.1 Illustration

A fair coin is tossed and a fair die is rolled.
Let

$$A = \{\text{coin shows heads}\}$$

and

$$B = \{\text{die shows } 6\}.$$

Then

$$\mathbb{P}(A) = \frac{1}{2}$$

and

$$\mathbb{P}(B) = \frac{1}{6}.$$

Since the coin toss has no influence on the die,

$$\mathbb{P}(A \cap B) = \frac{1}{12}.$$

Now

$$\mathbb{P}(A)\mathbb{P}(B) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}.$$

Therefore,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B),$$

so A and B are independent.

6.2 Mutually Exclusive versus Independent

These ideas must not be confused.

If two events are mutually exclusive,

$$A \cap B = \varnothing.$$

Therefore,

$$\mathbb{P}(A \cap B) = 0.$$

If both events have positive probability, then

$$\mathbb{P}(A)\mathbb{P}(B) > 0.$$

Hence they cannot be independent.

Mutually exclusive means:

they cannot happen together.

Independent means:

one happening does not affect the probability of the other.

They are fundamentally different ideas.

6.3 Illustration

A die is rolled.
Let

$$A = \{\text{result is } 2\}, \qquad B = \{\text{result is } 5\}.$$

Clearly,

$$A \cap B = \varnothing.$$

Thus

$$\mathbb{P}(A \cap B) = 0.$$

But

$$\mathbb{P}(A)\mathbb{P}(B) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}.$$

Since

$$0 \neq \frac{1}{36},$$

the events are not independent.

6.4 SOP for Testing Independence

Calculate

$$\mathbb{P}(A \cap B).$$

Separately calculate

$$\mathbb{P}(A)\mathbb{P}(B).$$

Then compare.
If

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B),$$

the events are independent.
Otherwise they are dependent.

7. Law of Total Probability

The law of total probability is used when an event can occur through several different mutually exclusive cases.

7.1 Partition of a Sample Space

Events

$$B_1, B_2, \dots, B_n$$

form a partition of the sample space if:

$$B_i \cap B_j = \varnothing \qquad (i \neq j)$$

and

$$B_1 \cup B_2 \cup \cdots \cup B_n = S.$$

That is:

- exactly one of the B_i's occurs;
- together they cover all possibilities.

7.2 Law

If

$$B_1, B_2, \ldots, B_n$$

form a partition, then for any event A,

$$\boxed{\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}$$

or explicitly,

$$\mathbb{P}(A) = \mathbb{P}(A \mid B_1)\mathbb{P}(B_1) + \mathbb{P}(A \mid B_2)\mathbb{P}(B_2) + \cdots + \mathbb{P}(A \mid B_n)\mathbb{P}(B_n).$$

7.3 Why it Works

The event A can be split into mutually exclusive pieces:

$$A = (A \cap B_1) \cup (A \cap B_2) \cup \cdots \cup (A \cap B_n).$$

Therefore,

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \cap B_i).$$

Using the multiplication rule,

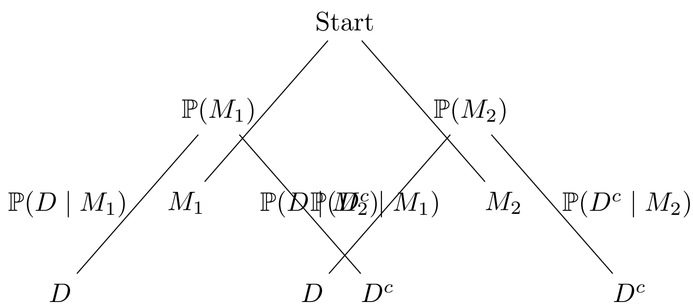
$$\mathbb{P}(A \cap B_i) = \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

Hence,

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

7.4 Tree Diagram

Suppose a factory has two machines M_1 and M_2.



For any complete path through a probability tree:

$$\boxed{\text{multiply along the branches}}$$

For alternative mutually exclusive paths leading to the same event:

$$\boxed{\text{add the path probabilities}}$$

7.5 Illustration

A factory uses two machines.
Machine A manufactures 60% of the components and machine B manufactures 40%.
Machine A has a defect rate of 2%, while machine B has a defect rate of 5%.
Find the probability that a randomly selected component is defective.
Let

$$D = \{\text{component is defective}\}.$$

We know

$$\mathbb{P}(A) = 0.60, \qquad \mathbb{P}(B) = 0.40,$$

$$\mathbb{P}(D \mid A) = 0.02, \qquad \mathbb{P}(D \mid B) = 0.05.$$

Using the law of total probability,

$$\mathbb{P}(D) = \mathbb{P}(D \mid A)\mathbb{P}(A) + \mathbb{P}(D \mid B)\mathbb{P}(B).$$

Therefore,

$$\mathbb{P}(D) = (0.02)(0.60) + (0.05)(0.40).$$

Thus

$$\mathbb{P}(D) = 0.012 + 0.020 = \boxed{0.032}.$$

Hence the overall defect rate is

$$\boxed{3.2\%}.$$

7.6 SOP

1. Identify the final event A whose probability is required.
2. Identify all mutually exclusive ways A can occur.
3. Define these cases as
$$B_1, B_2, \ldots, B_n.$$
4. Check that the cases cover the whole sample space.
5. For each case calculate
$$\mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$
6. Add all the path probabilities:
$$\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

8. Bayes' Theorem

Bayes' theorem reverses the direction of conditioning.

Suppose we know

$$\mathbb{P}(B \mid A)$$

but want

$$\mathbb{P}(A \mid B).$$

Bayes' theorem allows us to convert one into the other.

8.1 Derivation

From conditional probability,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Also,

$$\mathbb{P}(A \cap B) = \mathbb{P}(B \mid A)\mathbb{P}(A).$$

Therefore,

$$\boxed{\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}}$$

This is Bayes' theorem.

8.2 Terminology

In

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)},$$

the quantities are often described as follows:

$$\mathbb{P}(A) \quad \text{prior probability,}$$

$$\mathbb{P}(B \mid A) \quad \text{likelihood,}$$

$$\mathbb{P}(A \mid B) \quad \text{posterior probability.}$$

The prior represents our belief about A before observing B.
The posterior represents the updated probability after observing B.

8.3 General Bayes Formula

If

$$B_1, \dots, B_n$$

form a partition, then

$$\boxed{\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}}$$

The denominator is simply the law of total probability.

8.4 Illustration: Medical Test

Suppose a disease affects 1% of a population.
A test has:

$$\mathbb{P}(+ \mid D) = 0.95,$$

meaning a 95% sensitivity.
The false-positive rate is

$$\mathbb{P}(+ \mid D^c) = 0.05.$$

If a person tests positive, find the probability that the person actually has the disease.
We need

$$\mathbb{P}(D \mid +).$$

Bayes' theorem gives

$$\mathbb{P}(D \mid +) = \frac{\mathbb{P}(+ \mid D)\mathbb{P}(D)}{\mathbb{P}(+)}.$$

First calculate $\mathbb{P}(+)$ using total probability:

$$\mathbb{P}(+) = \mathbb{P}(+ \mid D)\mathbb{P}(D) + \mathbb{P}(+ \mid D^c)\mathbb{P}(D^c).$$

Now,

$$\mathbb{P}(D) = 0.01, \qquad \mathbb{P}(D^c) = 0.99.$$

Thus

$$\mathbb{P}(+) = (0.95)(0.01) + (0.05)(0.99).$$

$$\mathbb{P}(+) = 0.0095 + 0.0495 = 0.059.$$

Hence

$$\mathbb{P}(D \mid +) = \frac{(0.95)(0.01)}{0.059}.$$

Therefore,

$$\mathbb{P}(D \mid +) \approx \boxed{0.1610}.$$

So the probability is only approximately

$$\boxed{16.1\%}.$$

This occurs because the disease is rare. Even though the test is fairly accurate, false positives from the large healthy population are numerous.

8.5 Frequency Interpretation of the Same Example

Imagine 10 000 people.

Disease cases:

$$0.01(10000) = 100.$$

True positives:

$$0.95(100) = 95.$$

Healthy people:

$$9900.$$

False positives:

$$0.05(9900) = 495.$$

Total positive tests:

$$95 + 495 = 590.$$

Therefore,

$$\mathbb{P}(D \mid +) = \frac{95}{590} \approx 0.161.$$

This reproduces the Bayes result.

8.6 Illustration: Machines

Using the earlier factory:

$$\mathbb{P}(A) = 0.60, \qquad \mathbb{P}(B) = 0.40,$$

$$\mathbb{P}(D \mid A) = 0.02, \qquad \mathbb{P}(D \mid B) = 0.05.$$

We previously found

$$\mathbb{P}(D) = 0.032.$$

Suppose a component is known to be defective.
Find the probability that it was produced by machine B.
We require

$$\mathbb{P}(B \mid D).$$

Using Bayes' theorem,

$$\mathbb{P}(B \mid D) = \frac{\mathbb{P}(D \mid B)\mathbb{P}(B)}{\mathbb{P}(D)}.$$

Therefore,

$$\mathbb{P}(B \mid D) = \frac{(0.05)(0.40)}{0.032}.$$

Thus

$$\mathbb{P}(B \mid D) = \frac{0.020}{0.032} = \boxed{0.625}.$$

So 62.5% of defective components are expected to originate from machine B.

8.7 Bayes SOP

When the problem gives a probability in one direction and asks for the reverse direction:

1. Identify exactly what is required:
$$\mathbb{P}(\text{cause} \mid \text{observed evidence}).$$
2. Write Bayes' theorem:
$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$
3. Find the prior:
$$\mathbb{P}(A).$$
4. Find the likelihood:
$$\mathbb{P}(B \mid A).$$
5. If $\mathbb{P}(B)$ is not directly given, calculate it using the law of total probability.
6. Substitute carefully.
7. Interpret the posterior probability in words.

9. Random Variables and Probability Distributions

A probability distribution describes how probability is distributed over the possible numerical values of a random variable.

A **random variable** is a numerical quantity whose value depends on the outcome of a random experiment.
It is usually denoted by
$$X.$$

Examples:

- X = number of heads in 5 coin tosses,
- X = number of customers arriving in one hour,
- X = lifetime of a battery,
- X = height of a randomly selected person.

9.1 Discrete Random Variables

A discrete random variable takes countable values such as

$$0, 1, 2, 3, \ldots$$

Examples:

- number of defective products,
- number of goals,
- number of phone calls,
- number of successful trials.

The probability mass function is

$$p(x) = \mathbb{P}(X = x).$$

It must satisfy

$$p(x) \geq 0$$

and

$$\sum_x p(x) = 1.$$

9.2 Continuous Random Variables

A continuous random variable can take any real value over an interval.
Examples:

- time,
- height,
- temperature,
- weight.

For a continuous random variable,

$$\mathbb{P}(X = x) = 0.$$

Probabilities are calculated over intervals:

$$\mathbb{P}(a < X < b) = \int_a^b f(x)\,dx.$$

Here $f(x)$ is called the probability density function.

9.3 Expected Value

For a discrete random variable,

$$\boxed{\mathbb{E}[X] = \sum_x x\mathbb{P}(X = x)}$$

For a continuous random variable,

$$\boxed{\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x)\, dx}$$

The expected value represents the long-run average.

9.4 Variance

Variance measures spread around the mean.

$$\boxed{\mathrm{Var}(X) = \mathbb{E}[(X - \mu)^2]}$$

where

$$\mu = \mathbb{E}[X].$$

An extremely useful equivalent formula is

$$\boxed{\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2}$$

The standard deviation is

$$\boxed{\sigma = \sqrt{\mathrm{Var}(X)}}$$

10. Bernoulli Distribution

A Bernoulli random variable represents a single trial with exactly two possible outcomes. Examples:

- success or failure,
- defective or non-defective,
- heads or tails,
- purchase or no purchase,
- pass or fail.

Let

$$X = \begin{cases} 1, & \text{success,} \\ 0, & \text{failure.} \end{cases}$$

If

$$\mathbb{P}(X = 1) = p,$$

then

$$\mathbb{P}(X = 0) = 1 - p.$$

We write

$$\boxed{X \sim \text{Bernoulli}(p)}$$

Its PMF is

$$\boxed{\mathbb{P}(X = x) = p^x(1-p)^{1-x}, \qquad x \in \{0, 1\}}$$

10.1 Mean and Variance

For a Bernoulli random variable,

$$\boxed{\mathbb{E}[X] = p}$$

and

$$\boxed{\text{Var}(X) = p(1-p)}$$

10.2 Illustration

A component has probability 0.03 of being defective.
Let

$$X = \begin{cases} 1, & \text{component is defective,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$X \sim \text{Bernoulli}(0.03).$$

Therefore,

$$\mathbb{P}(X = 1) = 0.03$$

and

$$\mathbb{P}(X = 0) = 0.97.$$

The expected value is

$$\mathbb{E}[X] = 0.03.$$

The variance is

$$\text{Var}(X) = (0.03)(0.97) = \boxed{0.0291}.$$

11. Binomial Distribution

The binomial distribution counts the number of successes in several independent Bernoulli trials.
We write

$$\boxed{X \sim \text{Binomial}(n, p)}$$

where

$$n = \text{number of trials}$$

and

$$p = \text{probability of success on each trial.}$$

11.1 Conditions for a Binomial Model

A useful checklist is:

1. There is a fixed number n of trials.
2. Each trial has two outcomes: success or failure.
3. Trials are independent.
4. The success probability p is constant.
5. X counts the number of successes.

11.2 Deriving the Binomial Formula

Suppose exactly x successes occur.
One particular ordering containing x successes and $n - x$ failures has probability

$$p^x(1-p)^{n-x}.$$

But there are

$$\binom{n}{x}$$

ways to choose which x trials are successful.
Therefore,

$$\boxed{\mathbb{P}(X = x) = \binom{n}{x} p^x(1-p)^{n-x}}$$

for

$$x = 0, 1, \dots, n.$$

11.3 Mean and Variance

If

$$X \sim \text{Binomial}(n, p),$$

then

$$\boxed{\mathbb{E}[X] = np}$$

and

$$\boxed{\text{Var}(X) = np(1-p)}$$

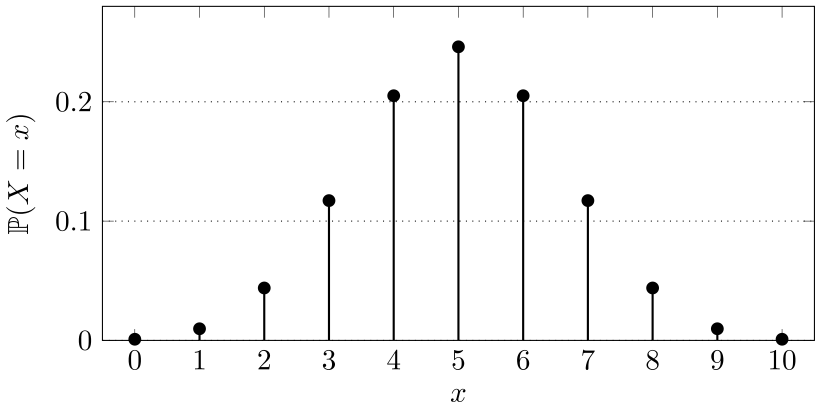
so

$$\boxed{\sigma = \sqrt{np(1-p)}}$$

11.4 Graph

For example, consider

$$X \sim \text{Binomial}(10, 0.5).$$



11.5 Illustration: Exact Probability

Suppose the probability that a customer purchases a product is 0.4.
Five independent customers are observed.
Find the probability that exactly 3 customers purchase the product.
Let

$$X \sim \text{Binomial}(5, 0.4).$$

We require

$$\mathbb{P}(X = 3).$$

Use

$$\mathbb{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}.$$

Therefore,

$$\mathbb{P}(X = 3) = \binom{5}{3} (0.4)^3 (0.6)^2.$$

Since

$$\binom{5}{3} = 10,$$

we obtain

$$\mathbb{P}(X = 3) = 10(0.064)(0.36).$$

Thus

$$\mathbb{P}(X = 3) = \boxed{0.2304}.$$

11.6 Illustration: At Least One

A machine produces a defective component with probability 0.02.
Ten independent components are inspected.
Find the probability of finding at least one defective component.
Let

$$X \sim \text{Binomial}(10, 0.02).$$

Directly calculating

$$\mathbb{P}(X \geq 1)$$

would require many terms.
Instead use the complement:

$$\mathbb{P}(X \geq 1) = 1 - \mathbb{P}(X = 0).$$

Now

$$\mathbb{P}(X = 0) = \binom{10}{0}(0.02)^0(0.98)^{10}.$$

Hence

$$\mathbb{P}(X = 0) = (0.98)^{10}.$$

Therefore,

$$\mathbb{P}(X \geq 1) = 1 - (0.98)^{10}.$$

Thus

$$\boxed{\mathbb{P}(X \geq 1) \approx 0.1829}$$

11.7 Useful Binomial Probability Forms

Exactly k:

$$\mathbb{P}(X = k).$$

At most k:

$$\mathbb{P}(X \leq k).$$

Less than k:

$$\mathbb{P}(X < k) = \mathbb{P}(X \leq k - 1).$$

At least k:

$$\mathbb{P}(X \geq k).$$

More than k:

$$\mathbb{P}(X > k) = \mathbb{P}(X \geq k+1).$$

At least one:

$$\boxed{\mathbb{P}(X \geq 1) = 1 - \mathbb{P}(X = 0)}$$

11.8 Binomial SOP

1. Confirm the binomial assumptions.
2. Identify
$$n, \qquad p, \qquad x.$$
3. Write
$$X \sim \text{Binomial}(n, p).$$
4. Translate the wording:
$$\text{exactly } k \to X = k,$$
$$\text{at most } k \to X \leq k,$$
$$\text{at least } k \to X \geq k.$$
5. For an exact value, use
$$\mathbb{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}.$$
6. Look for a complement before adding many terms.
7. Check whether the answer lies between 0 and 1.

12. Geometric Distribution

The geometric distribution models the number of independent Bernoulli trials required to obtain the **first success**.

Let

$$X = \text{trial number on which the first success occurs.}$$

Then

$$X = 1, 2, 3, \dots$$

and

$$\boxed{X \sim \text{Geometric}(p)}$$

12.1 PMF

For the first success to occur on trial k,

- the first $k-1$ trials must fail;
- the k-th trial must succeed.

Thus

$$\boxed{\mathbb{P}(X = k) = (1-p)^{k-1} p}$$

12.2 Mean and Variance

$$\boxed{\mathbb{E}[X] = \frac{1}{p}}$$

and

$$\boxed{\text{Var}(X) = \frac{1-p}{p^2}}$$

12.3 Tail Probability

The probability that more than k trials are required is

$$\boxed{\mathbb{P}(X > k) = (1-p)^k}$$

12.4 Memoryless Property

The geometric distribution is memoryless:

$$\boxed{\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)}$$

Past failures do not alter the success probability of future independent trials.

12.5 Illustration

A basketball player independently makes each free throw with probability

$$p = 0.7.$$

Find the probability that the player's first successful throw occurs on the fourth attempt. For the first success to occur on attempt 4, the sequence must be

$$F, F, F, S.$$

Therefore,

$$\mathbb{P}(X = 4) = (1 - 0.7)^3(0.7).$$

Thus

$$\mathbb{P}(X = 4) = (0.3)^3(0.7) = \boxed{0.0189}.$$

12.6 SOP

Use the geometric distribution when the question asks:

"How many trials until the first success?"

Then:

1. Identify p.
2. Identify the required trial number k.
3. Use
$$\mathbb{P}(X = k) = (1 - p)^{k-1}p.$$
4. For more than k trials, use
$$\mathbb{P}(X > k) = (1 - p)^k.$$

13. Poisson Distribution

The Poisson distribution models the number of events occurring in a fixed interval of time, distance, area or volume when events occur independently at an approximately constant average rate.

Examples include:

- phone calls received per minute,
- typing errors per page,

- accidents per month,
- defects per metre of material,
- customers arriving per hour.

We write

$$\boxed{X \sim \text{Poisson}(\lambda)}$$

where

$$\lambda = \text{expected number of events in the interval.}$$

13.1 PMF

$$\boxed{\mathbb{P}(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots}$$

13.2 Mean and Variance

For a Poisson distribution,

$$\boxed{\mathbb{E}[X] = \lambda}$$

and

$$\boxed{\text{Var}(X) = \lambda}$$

Therefore,

$$\boxed{\sigma = \sqrt{\lambda}}$$

13.3 Scaling the Rate

Suppose events occur at an average rate of 6 per hour.
For half an hour,

$$\lambda = 3.$$

For two hours,

$$\lambda = 12.$$

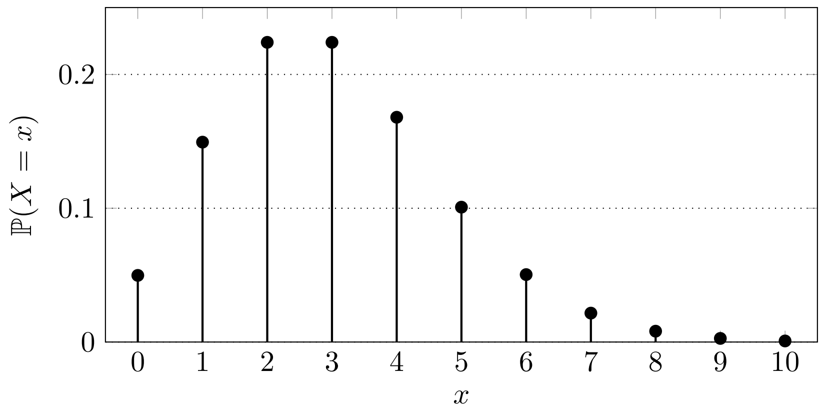
Always adjust λ to match the interval in the question.

13.4 Graph

For

$$X \sim \text{Poisson}(3),$$

the distribution has the approximate shape below.



13.5 Illustration

A help desk receives an average of 4 calls per hour.
Assume the number of calls follows a Poisson distribution.
Find the probability that exactly 2 calls arrive in the next hour.
Here,

$$\lambda = 4$$

and

$$x = 2.$$

Therefore,

$$\mathbb{P}(X = 2) = \frac{\mathrm{e}^{-4} 4^2}{2!}.$$

Thus

$$\mathbb{P}(X = 2) = 8\mathrm{e}^{-4}.$$

Hence

$$\boxed{\mathbb{P}(X = 2) \approx 0.1465}$$

13.6 Illustration: No Events

A server experiences an average of 3 failures per month.
Find the probability that there are no failures in a particular month.

$$X \sim \text{Poisson}(3).$$

We need

$$\mathbb{P}(X = 0).$$

Therefore,

$$\mathbb{P}(X = 0) = \frac{e^{-3}3^0}{0!}.$$

Since

$$3^0 = 1 \qquad \text{and} \qquad 0! = 1,$$

we obtain

$$\boxed{\mathbb{P}(X = 0) = e^{-3} \approx 0.0498.}$$

13.7 Poisson Approximation to Binomial

If

$$X \sim \text{Binomial}(n, p)$$

where n is large and p is small, the binomial distribution can often be approximated by

$$X \sim \text{Poisson}(\lambda)$$

with

$$\boxed{\lambda = np.}$$

This is especially useful for rare events.

13.8 SOP

1. Confirm that X counts events in an interval.
2. Identify the average event rate.
3. Convert the rate to the interval asked in the problem.
4. Set this value equal to λ.
5. Translate the wording into $X = x$, $X \leq x$, etc.
6. For exactly x, use
$$\mathbb{P}(X = x) = \frac{\mathrm{e}^{-\lambda}\lambda^x}{x!}.$$
7. Use complements for phrases such as "at least one" whenever convenient.

14. Continuous Uniform Distribution

A continuous random variable follows a uniform distribution if every interval of equal length within a given range has equal probability.
We write

$$\boxed{X \sim U(a, b)}$$

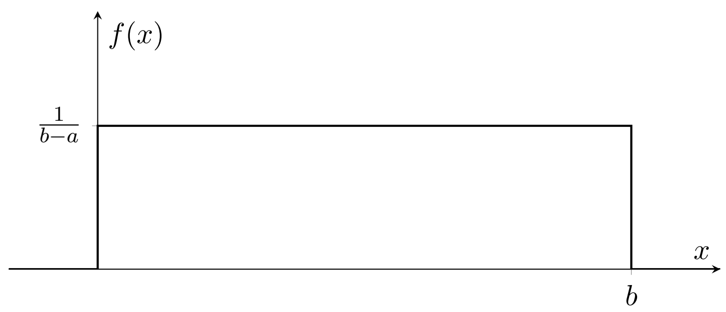
where X can take values between a and b.

14.1 PDF

The density is

$$\boxed{f(x) = \begin{cases} \dfrac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}}$$

The graph is a rectangle.



14.2 Interval Probabilities

If

$$a \leq c < d \leq b,$$

then

$$\mathbb{P}(c < X < d) = \frac{d-c}{b-a}.$$

Hence,

$$\boxed{\text{probability} = \frac{\text{desired interval length}}{\text{total interval length}}}$$

14.3 Mean and Variance

$$\boxed{\mathbb{E}[X] = \frac{a+b}{2}}$$

and

$$\boxed{\operatorname{Var}(X) = \frac{(b-a)^2}{12}}$$

14.4 CDF

The cumulative distribution function is

$$F(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \leq x \leq b, \\ 1, & x > b. \end{cases}$$

14.5 Illustration

A bus arrives at a random time uniformly between 0 and 20 minutes.
Let

$$X \sim U(0, 20).$$

Find the probability that the waiting time is between 5 and 12 minutes.
The total interval length is

$$20 - 0 = 20.$$

The desired interval length is

$$12 - 5 = 7.$$

Therefore,

$$\mathbb{P}(5 < X < 12) = \frac{7}{20} = \boxed{0.35}.$$

14.6 SOP

1. Identify the endpoints a and b.
2. Confirm that all positions inside the interval are equally likely.
3. Find the total length:
$$b - a.$$
4. Find the desired interval length.
5. Divide:
$$\mathbb{P}(c < X < d) = \frac{d - c}{b - a}.$$

15. Exponential Distribution

The exponential distribution is commonly used for waiting times.

Examples:

- waiting time until the next customer,
- lifetime of certain electronic components,
- waiting time until the next system failure,
- time between Poisson arrivals.

We write

$$\boxed{X \sim \text{Exponential}(\lambda)}$$

where

$$\lambda > 0$$

is the event rate.

15.1 PDF

$$\boxed{f(x) = \lambda e^{-\lambda x}, \qquad x \geq 0}$$

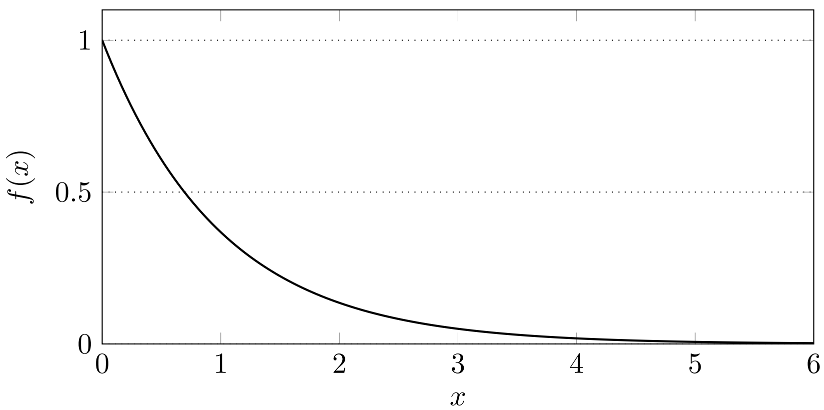
and

$$f(x) = 0$$

for $x < 0$.

15.2 Graph

For $\lambda = 1$:



The density is largest near 0 and decreases exponentially.

15.3 CDF

$$\boxed{F(x) = \mathbb{P}(X \leq x) = 1 - e^{-\lambda x}}$$

for $x \geq 0$.

15.4 Survival Probability

An especially useful formula is

$$\boxed{\mathbb{P}(X > x) = e^{-\lambda x}}$$

Therefore,

$$\mathbb{P}(X \leq x) = 1 - \mathrm{e}^{-\lambda x}.$$

For an interval,

$$\boxed{\mathbb{P}(a < X < b) = \mathrm{e}^{-\lambda a} - \mathrm{e}^{-\lambda b}}$$

15.5 Mean and Variance

$$\boxed{\mathbb{E}[X] = \frac{1}{\lambda}}$$

and

$$\boxed{\mathrm{Var}(X) = \frac{1}{\lambda^2}}$$

so

$$\boxed{\sigma = \frac{1}{\lambda}.}$$

15.6 Memoryless Property

The exponential distribution is memoryless:

$$\boxed{\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)}$$

This means that, under an exponential model, the probability of surviving an additional amount of time does not depend on how long the object has already survived.

15.7 Relationship with Poisson Distribution

The Poisson distribution counts **how many** events occur.

The exponential distribution measures **how long** we wait between events.

If events occur according to a Poisson process with rate λ, then the time between consecutive events follows an exponential distribution with the same rate λ.

15.8 Illustration

Customers arrive at a service counter at an average rate of 4 per hour.
Assume a Poisson process.
Let X be the waiting time in hours until the next customer.
Then

$$X \sim \text{Exponential}(4).$$

Find the probability that the next customer arrives after more than 15 minutes.
Convert 15 minutes to hours:

$$15 \text{ min} = \frac{1}{4} \text{ hour}.$$

We require

$$\mathbb{P}\left(X > \frac{1}{4}\right).$$

Using

$$\mathbb{P}(X > x) = \mathrm{e}^{-\lambda x},$$

we obtain

$$\mathbb{P}\left(X > \frac{1}{4}\right) = \mathrm{e}^{-4(1/4)}.$$

Therefore,

$$\mathbb{P}\left(X > \frac{1}{4}\right) = \mathrm{e}^{-1} \approx \boxed{0.3679}.$$

15.9 SOP

1. Confirm that the variable represents waiting time or lifetime.
2. Identify the rate λ.
3. Ensure time units match the rate units.
4. For "more than x", use
$$\mathbb{P}(X > x) = \mathrm{e}^{-\lambda x}.$$
5. For "less than x", use
$$\mathbb{P}(X < x) = 1 - \mathrm{e}^{-\lambda x}.$$
6. For an interval use
$$\mathbb{P}(a < X < b) = \mathrm{e}^{-\lambda a} - \mathrm{e}^{-\lambda b}.$$

16. Normal Distribution

The normal distribution is one of the most important probability distributions in statistics.

It is appropriate for many quantities formed through the combined effect of numerous small random influences.

Examples often include:

- measurement errors,
- biological measurements,
- heights,
- manufacturing dimensions,
- examination scores under suitable conditions.

We write

$$\boxed{X \sim N(\mu, \sigma^2)}$$

where

$$\mu = \text{mean}$$

and

$$\sigma^2 = \text{variance}.$$

16.1 Probability Density Function

The normal density is

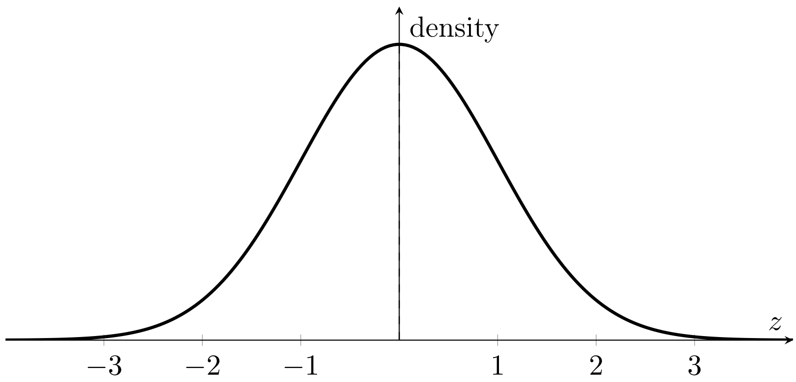
$$\boxed{f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]}$$

for

$$-\infty < x < \infty.$$

16.2 Normal Curve

For the standard normal distribution:



16.3 Properties

A normal distribution:

1. is bell-shaped;
2. is symmetric about μ;
3. has
$$\text{mean} = \text{median} = \text{mode} = \mu;$$
4. has total area 1;
5. approaches the horizontal axis but never reaches it;
6. is completely determined by μ and σ.

16.4 Role of the Mean

The mean determines the centre of the distribution.

Increasing μ shifts the entire curve to the right.

Decreasing μ shifts it to the left.

16.5 Role of the Standard Deviation

The standard deviation controls spread.

A small σ gives a narrow, tall curve.

A large σ gives a wide, flat curve.

16.6 The Standard Normal Distribution

A standard normal random variable has

$$\mu = 0$$

and

$$\sigma = 1.$$

It is written

$$\boxed{Z \sim N(0,1).}$$

16.7 Standardisation

If

$$X \sim N(\mu, \sigma^2),$$

then define

$$\boxed{Z = \frac{X - \mu}{\sigma}.}$$

This converts X into a standard normal random variable.

The resulting Z-value tells us how many standard deviations an observation lies above or below the mean.

For example,

$$Z = 2$$

means two standard deviations above the mean.

$$Z = -1.5$$

means 1.5 standard deviations below the mean.

16.8 The 68–95–99.7 Rule

Approximately:

$$68\%$$

of observations lie within

$$\mu \pm \sigma.$$

Approximately:

$$95\%$$

lie within

$$\mu \pm 2\sigma.$$

Approximately:

$$99.7\%$$

lie within

$$\mu \pm 3\sigma.$$

Thus,

$$\mathbb{P}(\mu - \sigma < X < \mu + \sigma) \approx 0.68,$$

$$\mathbb{P}(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.95,$$

$$\mathbb{P}(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.997.$$

16.9 Some Useful Standard Normal Values

Let

$$\Phi(z) = \mathbb{P}(Z \leq z).$$

Some useful approximate values are:

z	$\Phi(z)$
0.00	0.5000
0.50	0.6915
1.00	0.8413
1.28	0.8997
1.645	0.9500
1.96	0.9750
2.00	0.9772
2.33	0.9901
2.58	0.9951
3.00	0.9987

By symmetry,

$$\boxed{\Phi(-z) = 1 - \Phi(z).}$$

16.10 Illustration: Probability below a value

Suppose examination scores are normally distributed with

$$\mu = 70$$

and

$$\sigma = 10.$$

Find the probability that a randomly selected student scores below 80.
Let

$$X \sim N(70, 10^2).$$

Standardise:

$$Z = \frac{80 - 70}{10} = 1.$$

Therefore,

$$\mathbb{P}(X < 80) = \mathbb{P}(Z < 1).$$

From the standard normal table,

$$\mathbb{P}(Z < 1) = 0.8413.$$

Hence

$$\boxed{\mathbb{P}(X < 80) = 0.8413.}$$

16.11 Illustration: Probability between two values

Suppose

$$X \sim N(100, 15^2).$$

Find

$$\mathbb{P}(85 < X < 115).$$

For $X = 85$,

$$Z_1 = \frac{85 - 100}{15} = -1.$$

For $X = 115$,

$$Z_2 = \frac{115 - 100}{15} = 1.$$

Thus

$$\mathbb{P}(85 < X < 115) = \mathbb{P}(-1 < Z < 1).$$

Using the table,

$$\mathbb{P}(Z < 1) = 0.8413$$

and

$$\mathbb{P}(Z < -1) = 0.1587.$$

Therefore,

$$\mathbb{P}(-1 < Z < 1) = 0.8413 - 0.1587.$$

Hence,

$$\boxed{\mathbb{P}(85 < X < 115) = 0.6826.}$$

16.12 Illustration: Finding a percentile

Scores follow

$$X \sim N(500, 100^2).$$

Find the score separating the highest 2.5% from the remaining 97.5%.
We need x such that

$$\mathbb{P}(X < x) = 0.975.$$

From the standard normal table,

$$z = 1.96.$$

Using

$$z = \frac{x - \mu}{\sigma},$$

we get

$$1.96 = \frac{x - 500}{100}.$$

Therefore,

$$x - 500 = 196,$$

so

$$\boxed{x = 696}.$$

16.13 Normal Distribution SOP

1. Identify
$$\mu, \qquad \sigma.$$
2. Write
$$X \sim N(\mu, \sigma^2).$$
3. Translate the required probability into an inequality involving X.
4. Standardise every boundary:
$$z = \frac{x-\mu}{\sigma}.$$
5. Sketch the normal curve if the direction is unclear.
6. Use the standard normal CDF/table.
7. Remember:
$$\mathbb{P}(Z > z) = 1 - \Phi(z).$$
8. For an interval:
$$\mathbb{P}(a < X < b) = \Phi(z_b) - \Phi(z_a).$$
9. Check whether the result is sensible from the sketch.

17. How to Choose the Correct Distribution

A major source of difficulty is not performing the calculation but deciding which distribution applies.

Distribution	**Main question**	**Typical clue**
Bernoulli	What happens in one success/failure trial?	One trial, two outcomes
Binomial	How many successes occur in n trials?	Fixed n, constant p, independent trials
Geometric	When does the first success occur?	Number of trials until first success
Poisson	How many events occur in an interval?	Count per unit time, area, length or volume
Uniform	Where does a value lie in an equally likely interval?	Constant density over a finite interval
Exponential	How long until the next event?	Waiting time between Poisson events
Normal	How is a continuous quantity distributed around a mean?	Bell-shaped continuous measurements

17.1 Decision Procedure

Ask the following questions in order.

1. Is the variable **discrete** or **continuous**?
2. If discrete:
   - One success/failure trial?
     $\rightarrow$ Bernoulli
   - Number of successes in fixed n trials?
     $\rightarrow$ Binomial
   - Number of trials until first success?
     $\rightarrow$ Geometric
   - Number of arrivals/events in an interval?
     $\rightarrow$ Poisson
3. If continuous:
   - Constant density over a bounded interval?
     $\rightarrow$ Uniform
   - Waiting time until an event?
     $\rightarrow$ Exponential
   - Symmetric bell-shaped measurement?
     $\rightarrow$ Normal

18. Connections Between the Distributions

The distributions are not isolated topics.
They are closely connected.

18.1 Bernoulli to Binomial

If

$$X_1, X_2, \ldots, X_n$$

are independent Bernoulli random variables with probability p, then

$$X = X_1 + \cdots + X_n$$

counts the number of successes.
Therefore,

$$X \sim \text{Binomial}(n, p).$$

18.2 Bernoulli to Geometric

Repeated Bernoulli trials can also produce a geometric random variable if we ask:

How long until the first success?

Thus:

$$\text{fixed number of trials} \quad \to \quad \text{Binomial},$$

whereas

$$\text{trials until first success} \quad \to \quad \text{Geometric}.$$

18.3 Poisson and Exponential

Within a Poisson process,

$$\text{number of events in time } t$$

is Poisson distributed, whereas

$$\text{waiting time between events}$$

is exponentially distributed.
This distinction is extremely important.

18.4 Binomial and Poisson Approximation

For large n and small p,

$$\text{Binomial}(n, p)$$

can be approximated using

$$\text{Poisson}(np).$$

19. Mixed Worked Illustrations

19.1 Worked Illustration 1

Suppose

$$\mathbb{P}(A) = 0.7, \qquad \mathbb{P}(B) = 0.5, \qquad \mathbb{P}(A \cap B) = 0.4.$$

Find

$$\mathbb{P}(A \mid B).$$

By definition,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Therefore,

$$\mathbb{P}(A \mid B) = \frac{0.4}{0.5} = \boxed{0.8}.$$

19.2 Worked Illustration 2

Suppose

$$\mathbb{P}(A) = 0.4, \qquad \mathbb{P}(B) = 0.3,$$

and

$$\mathbb{P}(A \cap B) = 0.12.$$

Test for independence.
Calculate

$$\mathbb{P}(A)\mathbb{P}(B) = (0.4)(0.3) = 0.12.$$

Since

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B),$$

the events are

$$\boxed{\text{independent}}.$$

19.3 Worked Illustration 3

Three suppliers provide 20%, 30%, and 50% of a company's components.
Their defect rates are respectively

$$1\%, \qquad 2\%, \qquad 4\%.$$

Let the suppliers be S_1, S_2, S_3, and let D represent a defective component.
First calculate

$$\mathbb{P}(D).$$

Using total probability,

$$\mathbb{P}(D) = (0.20)(0.01) + (0.30)(0.02) + (0.50)(0.04).$$

Therefore,

$$\mathbb{P}(D) = 0.002 + 0.006 + 0.020 = \boxed{0.028}.$$

Now suppose the component is defective.
Find the probability that it came from supplier S_3.
Using Bayes,

$$\mathbb{P}(S_3 \mid D) = \frac{\mathbb{P}(D \mid S_3)\mathbb{P}(S_3)}{\mathbb{P}(D)}.$$

Thus

$$\mathbb{P}(S_3 \mid D) = \frac{(0.04)(0.50)}{0.028}.$$

Therefore,

$$\mathbb{P}(S_3 \mid D) = \frac{0.020}{0.028} \approx \boxed{0.7143}.$$

19.4 Worked Illustration 4

A website visitor has probability 0.2 of making a purchase.
Assume visitors behave independently.
Part A: What is the probability that exactly 2 out of the next 5 visitors purchase?
This is binomial:

$$X \sim \text{Binomial}(5, 0.2).$$

Therefore,

$$\mathbb{P}(X = 2) = \binom{5}{2}(0.2)^2(0.8)^3.$$

Thus

$$\mathbb{P}(X = 2) = 10(0.04)(0.512) = \boxed{0.2048}.$$

Part B: What is the probability that the first purchase occurs on the fifth visitor?
This is geometric:

$$Y \sim \text{Geometric}(0.2).$$

Therefore,

$$\mathbb{P}(Y = 5) = (0.8)^4(0.2).$$

Hence

$$\boxed{\mathbb{P}(Y = 5) = 0.08192.}$$

The same probability p appears in both models, but the random variables ask fundamentally different questions.

19.5 Worked Illustration 5

A data centre receives an average of 6 critical alerts per hour.
Part A: Find the probability of exactly 3 alerts in one hour.
Let

$$X \sim \text{Poisson}(6).$$

Then

$$\mathbb{P}(X = 3) = \frac{e^{-6}6^3}{3!}.$$

Thus

$$\mathbb{P}(X = 3) = 36e^{-6} \approx \boxed{0.0892}.$$

Part B: Find the probability that the waiting time until the next alert exceeds 20 minutes.
Let T denote waiting time in hours.
Then

$$T \sim \text{Exponential}(6).$$

Since

$$20 \text{ minutes} = \frac{1}{3} \text{ hour},$$

we obtain

$$\mathbb{P}\left(T > \frac{1}{3}\right) = e^{-6(1/3)} = e^{-2}.$$

Therefore,

$$\boxed{\mathbb{P}\left(T > \frac{1}{3}\right) \approx 0.1353.}$$

20. Common Errors

Do not assume

$$\mathbb{P}(A \mid B) = \mathbb{P}(B \mid A).$$

Always identify the event after the vertical bar.

Mutually exclusive events with positive probability are dependent.
If A occurs, it guarantees that B did not occur.

If the rate is 8 per hour but the question asks about 30 minutes, use

$$\lambda = 4,$$

not 8.

Poisson:

number of events.

Exponential:

waiting time.

A binomial model requires independent trials with a constant success probability.
Sampling without replacement from a small population generally violates this assumption.

The correct formula is

$$Z = \frac{X - \mu}{\sigma},$$

not

$$\frac{X - \mu}{\sigma^2}.$$

21. Practice Questions

Attempt these before reading the answers.

21.1 Basic Probability

Q1. A fair die is rolled. Find the probability of obtaining:

(a) an odd number,

(b) a number greater than 2,

(c) an odd number greater than 2.

Q2. Two fair coins are tossed. Find the probability of:

(a) exactly one head,

(b) at least one head,

(c) no heads.

Q3. Suppose

$$\mathbb{P}(A) = 0.6, \quad \mathbb{P}(B) = 0.5, \quad \mathbb{P}(A \cap B) = 0.3.$$

Find

$$\mathbb{P}(A \cup B).$$

Q4. Suppose

$$\mathbb{P}(A) = 0.72.$$

Find

$$\mathbb{P}(A^c).$$

21.2 Conditional Probability

Q5. Suppose

$$\mathbb{P}(A) = 0.5, \quad \mathbb{P}(B) = 0.4, \quad \mathbb{P}(A \cap B) = 0.2.$$

Find

$$\mathbb{P}(A \mid B).$$

Q6. In a class of 80 students, 50 know Python, 30 know C++, and 20 know both.

Given that a student knows C++, find the probability that the student also knows Python.

Q7. A card is selected from a standard deck. Given that it is a face card, find the probability that it is a king.

Q8. A die is rolled. Given that the result is even, find the probability that it is greater than 3.

21.3 Independence

Q9. Suppose

$$\mathbb{P}(A) = 0.2, \quad \mathbb{P}(B) = 0.5, \quad \mathbb{P}(A \cap B) = 0.1.$$

Are A and B independent?

Q10. Suppose

$$\mathbb{P}(A) = 0.3, \quad \mathbb{P}(B) = 0.4, \quad \mathbb{P}(A \cap B) = 0.08.$$

Are A and B independent?

Q11. A coin is tossed twice.

Let

$$A = \{\text{first toss is heads}\}$$

and

$$B = \{\text{second toss is heads}\}.$$

Show that A and B are independent.

21.4 Total Probability and Bayes

Q12. A company obtains 70% of its components from supplier A and 30% from supplier B.

Supplier A has a defect rate of 1%, while supplier B has a defect rate of 3%.

Find the probability that a randomly selected component is defective.

Q13. Using the previous question, if a component is defective, find the probability that it came from supplier B.

Q14. A disease affects 2% of a population.

A test detects the disease with probability 0.90 when the disease is present. The false-positive rate is 0.04.

Find the probability that a person who tests positive actually has the disease.

Q15. Three machines produce 25%, 35%, and 40% of all products. Their defect rates are respectively 1%, 2%, and 5%.

Find:

(a) the overall defect probability;

(b) the probability that a defective product came from the third machine.

21.5 Bernoulli and Binomial

Q16. A student answers a true/false question randomly. Define a Bernoulli random variable for a correct answer and state p, the mean and the variance.

Q17. A fair coin is tossed 8 times. Find the probability of exactly 5 heads.

Q18. A component has defect probability 0.1. Ten independent components are examined. Find:

(a) the probability of exactly 2 defectives;

(b) the probability of no defectives;

(c) the probability of at least one defective.

Q19. If

$$X \sim \text{Binomial}(20, 0.3),$$

find:

(a) $\mathbb{E}[X]$,

(b) $\text{Var}(X)$,

(c) the standard deviation.

Q20. A multiple-choice examination contains 10 questions, each with 4 choices and one correct choice. A student guesses every answer. Find the probability of exactly 3 correct answers.

21.6 Geometric

Q21. A fair coin is repeatedly tossed. Find the probability that the first head occurs on toss 5.

Q22. A trial succeeds with probability 0.25. Find the probability that more than 4 trials are required to obtain the first success.

Q23. If

$$X \sim \text{Geometric}(0.2),$$

find

$$\mathbb{E}[X]$$

and

$$\text{Var}(X).$$

21.7 Poisson

Q24. A website receives an average of 5 requests per second. Assuming a Poisson model, find the probability of exactly 3 requests in one second.

Q25. A machine experiences an average of 2 failures per month. Find the probability of no failures in a month.

Q26. Cars arrive at a checkpoint at an average rate of 12 per hour. Find the probability of exactly 4 cars arriving in 20 minutes.

Q27. An item has defect probability 0.002. A batch contains 1000 independent items.

Use a Poisson approximation to estimate the probability of exactly 3 defectives.

21.8 Uniform

Q28. Suppose

$$X \sim U(0, 10).$$

Find

$$\mathbb{P}(3 < X < 7).$$

Q29. Suppose

$$X \sim U(20, 50).$$

Find:

(a) $\mathbb{E}[X]$,

(b) $\text{Var}(X)$,

(c) $\mathbb{P}(X < 30)$.

21.9 Exponential

Q30. The lifetime of a component is exponentially distributed with mean 5 years.

Find the probability that it lasts more than 3 years.

Q31. Customers arrive at an average rate of 6 per hour.

Find the probability that the next customer arrives within 10 minutes.

Q32. If

$$X \sim \text{Exponential}(0.5),$$

find:

(a) $\mathbb{E}[X]$,

(b) $\text{Var}(X)$,

(c) $\mathbb{P}(X > 4)$.

21.10 Normal

Q33. Suppose

$$X \sim N(50, 10^2).$$

Find the Z-score corresponding to $X = 70$.

Q34. Suppose

$$X \sim N(100, 20^2).$$

Find

$$\mathbb{P}(X < 120).$$

Use

$$\Phi(1) = 0.8413.$$

Q35. Suppose

$$X \sim N(60, 5^2).$$

Find

$$\mathbb{P}(55 < X < 65).$$

Use

$$\Phi(1) = 0.8413.$$

Q36. Heights are normally distributed with mean 170 cm and standard deviation 8 cm.

Find the height corresponding to a Z-score of 1.5.

Q37. A variable is normally distributed with

$$\mu = 40, \qquad \sigma = 4.$$

Approximately what percentage of observations lie between 32 and 48?

22. Answers to Practice Questions

Q1.

$$(a)\ \frac{1}{2}, \qquad (b)\ \frac{2}{3}, \qquad (c)\ \frac{1}{3}.$$

Q2.

$$(a)\ \frac{1}{2}, \qquad (b)\ \frac{3}{4}, \qquad (c)\ \frac{1}{4}.$$

Q3.

$$\mathbb{P}(A \cup B) = 0.6 + 0.5 - 0.3 = \boxed{0.8}.$$

Q4.

$$\mathbb{P}(A^c) = 1 - 0.72 = \boxed{0.28}.$$

Q5.

$$\mathbb{P}(A \mid B) = \frac{0.2}{0.4} = \boxed{0.5}.$$

Q6.

$$\mathbb{P}(\text{Python} \mid \text{C++}) = \frac{20}{30} = \boxed{\frac{2}{3}}.$$

Q7. There are 12 face cards and 4 kings, so

$$\boxed{\frac{4}{12} = \frac{1}{3}}.$$

Q8. Given an even outcome, the reduced sample space is

$$\{2, 4, 6\}.$$

The favourable outcomes are $4, 6$.

Thus

$$\boxed{\frac{2}{3}}.$$

Q9.

$$\mathbb{P}(A)\mathbb{P}(B) = (0.2)(0.5) = 0.1.$$

Since this equals $\mathbb{P}(A \cap B)$,

$$\boxed{\text{independent}}.$$

Q10.

$$\mathbb{P}(A)\mathbb{P}(B) = (0.3)(0.4) = 0.12.$$

But

$$\mathbb{P}(A \cap B) = 0.08.$$

Therefore,

$$\boxed{\text{not independent}}.$$

Q11.

$$\mathbb{P}(A) = \frac{1}{2}, \qquad \mathbb{P}(B) = \frac{1}{2},$$

and

$$\mathbb{P}(A \cap B) = \frac{1}{4}.$$

Since

$$\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2},$$

they are independent.

Q12.

$$\mathbb{P}(D) = (0.70)(0.01) + (0.30)(0.03)$$

$$= 0.007 + 0.009 = \boxed{0.016}.$$

Q13.

$$\mathbb{P}(B \mid D) = \frac{(0.03)(0.30)}{0.016} = \boxed{0.5625}.$$

Q14.

$$\mathbb{P}(D) = 0.02, \qquad \mathbb{P}(D^c) = 0.98.$$

The total positive probability is

$$\mathbb{P}(+) = (0.90)(0.02) + (0.04)(0.98).$$

Thus

$$\mathbb{P}(+) = 0.018 + 0.0392 = 0.0572.$$

Therefore,

$$\mathbb{P}(D \mid +) = \frac{0.018}{0.0572} \approx \boxed{0.3147}.$$

Q15.

$$\mathbb{P}(D) = (0.25)(0.01) + (0.35)(0.02) + (0.40)(0.05)$$

$$= 0.0025 + 0.007 + 0.020 = \boxed{0.0295}.$$

Then

$$\mathbb{P}(M_3 \mid D) = \frac{0.40(0.05)}{0.0295} = \boxed{0.6780 \text{ approximately}}.$$

Q16. For a random true/false guess,

$$p = \frac{1}{2}.$$

Hence

$$\mathbb{E}[X] = \frac{1}{2}$$

and

$$\mathrm{Var}(X) = \frac{1}{2}\left(1 - \frac{1}{2}\right) = \boxed{\frac{1}{4}}.$$

Q17.

$$\mathbb{P}(X = 5) = \binom{8}{5}\left(\frac{1}{2}\right)^5\left(\frac{1}{2}\right)^3$$

$$= \frac{\binom{8}{5}}{2^8} = \frac{56}{256} = \boxed{0.21875}.$$

Q18. Here

$$X \sim \text{Binomial}(10, 0.1).$$

For exactly 2,

$$\mathbb{P}(X = 2) = \binom{10}{2}(0.1)^2(0.9)^8 \approx \boxed{0.1937}.$$

For none,

$$\mathbb{P}(X = 0) = (0.9)^{10} \approx \boxed{0.3487}.$$

For at least one,

$$\mathbb{P}(X \geq 1) = 1 - (0.9)^{10} \approx \boxed{0.6513}.$$

Q19.

$$\mathbb{E}[X] = np = 20(0.3) = \boxed{6}.$$

$$\mathrm{Var}(X) = 20(0.3)(0.7) = \boxed{4.2}.$$

$$\sigma = \sqrt{4.2} \approx \boxed{2.049}.$$

Q20. Here

$$n = 10, \qquad p = \frac{1}{4}.$$

Therefore,

$$\mathbb{P}(X = 3) = \binom{10}{3} \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^7.$$

Numerically,

$$\boxed{\mathbb{P}(X = 3) \approx 0.2503}.$$

Q21.

$$\mathbb{P}(X = 5) = \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right) = \boxed{\frac{1}{32}}.$$

Q22.

$$\mathbb{P}(X > 4) = (1 - 0.25)^4 = (0.75)^4 = \boxed{0.3164}.$$

Q23. For $p = 0.2$,

$$\mathbb{E}[X] = \frac{1}{0.2} = \boxed{5}.$$

$$\mathrm{Var}(X) = \frac{0.8}{(0.2)^2} = \boxed{20}.$$

Q24.

$$\mathbb{P}(X = 3) = \frac{\mathrm{e}^{-5} 5^3}{3!}.$$

Thus

$$\boxed{\mathbb{P}(X = 3) \approx 0.1404}.$$

Q25.

$$\mathbb{P}(X = 0) = \mathrm{e}^{-2} \approx \boxed{0.1353}.$$

Q26. The average is 12 per hour.

Twenty minutes is one-third of an hour.

Thus

$$\lambda = 12 \left(\frac{1}{3}\right) = 4.$$

Therefore,

$$\mathbb{P}(X = 4) = \frac{\mathrm{e}^{-4} 4^4}{4!} \approx \boxed{0.1954}.$$

Q27. For the Poisson approximation,

$$\lambda = np = 1000(0.002) = 2.$$

Thus

$$\mathbb{P}(X = 3) \approx \frac{e^{-2}2^3}{3!} \approx \boxed{0.1804}.$$

Q28.

$$\mathbb{P}(3 < X < 7) = \frac{7-3}{10-0} = \boxed{0.4}.$$

Q29. For

$$X \sim U(20, 50),$$

$$\mathbb{E}[X] = \frac{20+50}{2} = \boxed{35}.$$

$$\mathrm{Var}(X) = \frac{(50-20)^2}{12} = \frac{900}{12} = \boxed{75}.$$

Also,

$$\mathbb{P}(X < 30) = \frac{30-20}{50-20} = \boxed{\frac{1}{3}}.$$

Q30. The mean is

$$\frac{1}{\lambda} = 5.$$

Therefore,

$$\lambda = 0.2.$$

Hence

$$\mathbb{P}(X > 3) = e^{-0.2(3)} = e^{-0.6} \approx \boxed{0.5488}.$$

Q31. Ten minutes is

$$\frac{1}{6} \text{ hour.}$$

Therefore,

$$\mathbb{P}\left(X < \frac{1}{6}\right) = 1 - e^{-6(1/6)}$$

$$= 1 - e^{-1} \approx \boxed{0.6321}.$$

Q32. If

$$\lambda = 0.5,$$

then

$$\mathbb{E}[X] = \frac{1}{0.5} = \boxed{2}.$$

$$\mathrm{Var}(X) = \frac{1}{(0.5)^2} = \boxed{4}.$$

Also,

$$\mathbb{P}(X > 4) = \mathrm{e}^{-0.5(4)} = \mathrm{e}^{-2} \approx \boxed{0.1353}.$$

Q33.

$$Z = \frac{70 - 50}{10} = \boxed{2}.$$

Q34.

$$Z = \frac{120 - 100}{20} = 1.$$

Therefore,

$$\mathbb{P}(X < 120) = \Phi(1) = \boxed{0.8413}.$$

Q35. For 55,

$$Z_1 = \frac{55 - 60}{5} = -1.$$

For 65,

$$Z_2 = \frac{65 - 60}{5} = 1.$$

Thus

$$\mathbb{P}(55 < X < 65) = \mathbb{P}(-1 < Z < 1).$$

Hence

$$= 0.8413 - 0.1587 = \boxed{0.6826}.$$

Q36. Using

$$z = \frac{x - \mu}{\sigma},$$

we have

$$1.5 = \frac{x - 170}{8}.$$

Therefore,

$$x - 170 = 12$$

and

$$\boxed{x = 182 \text{ cm}}.$$

Q37. The interval

$$32 < X < 48$$

is

$$40 - 2(4) < X < 40 + 2(4).$$

Thus it lies within two standard deviations of the mean.

By the empirical rule,

$$\boxed{\text{approximately } 95\%}.$$

23. Formula Sheet

Basic Probability

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$$

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

Conditional Probability

$$\boxed{\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}}$$

Multiplication Rule

$$\boxed{\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A)}$$

Independence

$$\boxed{\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)}$$

Law of Total Probability

$$\boxed{\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}$$

Bayes' Theorem

$$\boxed{\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}}$$

and

$$\boxed{\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}}$$

Bernoulli

$$\mathbb{P}(X = x) = p^x(1-p)^{1-x}$$

$$\mathbb{E}[X] = p$$

$$\mathrm{Var}(X) = p(1-p)$$

Binomial

$$\boxed{\mathbb{P}(X = x) = \binom{n}{x} p^x(1-p)^{n-x}}$$

$$\mathbb{E}[X] = np$$

$$\mathrm{Var}(X) = np(1-p)$$

Geometric

$$\boxed{\mathbb{P}(X = k) = (1-p)^{k-1}p}$$

$$\mathbb{P}(X > k) = (1-p)^k$$

$$\mathbb{E}[X] = \frac{1}{p}$$

$$\mathrm{Var}(X) = \frac{1-p}{p^2}$$

Poisson

$$\boxed{\mathbb{P}(X = x) = \frac{\mathrm{e}^{-\lambda}\lambda^x}{x!}}$$

$$\mathbb{E}[X] = \lambda$$

$$\mathrm{Var}(X) = \lambda$$

Uniform

$$f(x) = \frac{1}{b-a}, \qquad a \leq x \leq b$$

$$\mathbb{P}(c < X < d) = \frac{d-c}{b-a}$$

$$\mathbb{E}[X] = \frac{a+b}{2}$$

$$\mathrm{Var}(X) = \frac{(b-a)^2}{12}$$

Exponential

$$f(x) = \lambda\mathrm{e}^{-\lambda x}$$

$$\mathbb{P}(X > x) = \mathrm{e}^{-\lambda x}$$

$$\mathbb{P}(X \leq x) = 1 - \mathrm{e}^{-\lambda x}$$

$$\mathbb{E}[X] = \frac{1}{\lambda}$$

$$\mathrm{Var}(X) = \frac{1}{\lambda^2}$$

Normal

$$X \sim N(\mu, \sigma^2)$$

$$\boxed{Z = \frac{X-\mu}{\sigma}}$$

$$\Phi(z) = \mathbb{P}(Z \leq z)$$

$$\Phi(-z) = 1 - \Phi(z)$$

24. Final Problem-Solving Checklist

Before solving any probability problem:

1. Write down exactly what is random.
2. Define the required event or random variable.
3. Translate words into mathematical notation.
4. Ask whether the problem involves:
    - ordinary probability,
    - conditioning,
    - independence,
    - total probability,
    - Bayes' theorem,
    - or a probability distribution.
5. If conditioning appears, identify the reduced sample space.
6. If several mutually exclusive paths lead to one outcome, consider total probability.
7. If the question asks for the probability of a possible cause after observing an outcome, consider Bayes' theorem.
8. If a distribution is required, first determine whether the variable is discrete or continuous.
9. Identify all parameters before substituting into a formula.
10. Convert units before performing the calculation.
11. Look for a complement whenever the phrase "at least one" appears.
12. Draw a probability tree for multi-stage conditional problems.
13. Draw a normal curve before evaluating normal probabilities.
14. Check:
$$0 \leq \mathbb{P}(A) \leq 1.$$
15. Finally, interpret the numerical answer in the context of the original problem.

End of Unit 4